DL Practical 1

Boston Housing Price Prediction using Linear Regression & Deep Neural Network

Aim

Predict Boston house prices using Linear Regression with Deep Learning concepts.

Concept Explanation

This practical is based on **supervised learning**.

The model learns the relationship between:

- Input features (crime rate, rooms, tax, etc.)
- Output value (house price)

The dataset contains:

- 506 rows
- 13 input features
- 1 target column (Price)

Important Features in Dataset

Feature	Meaning
CRIM	Crime rate
RM	Number of rooms
AGE	Old houses proportion
TAX	Property tax
PTRATIO	Student-teacher ratio
LSTAT	Lower income population percentage

More rooms generally increase price, while crime rate lowers price.

What is Linear Regression?

Linear Regression predicts output using a straight-line equation.

The model tries to find:

$$y = \theta_1 + \theta_2 x$$

Where:

- y = predicted house price
- x = input feature
- θ1 = intercept
- θ2 = slope/weight

Working of the Practical

Step 1: Load Dataset

Boston Housing dataset is loaded using sklearn.

Step 2: Convert to DataFrame

Data is converted into pandas dataframe for easy handling.

Step 3: Split Dataset

Dataset divided into:

- Training data (80%)
- Testing data (20%)

Purpose:

- Train on one set
- Test on unseen data

Model Training

LinearRegression() model is used.

The model:

1. Learns patterns from training data
2. Finds best-fit line
3. Predicts house prices

Cost Function

Cost function calculates the difference between actual output and predicted output and helps evaluate model performance.

The Cost Function measures how far the predicted prices are from the actual prices.

In this practical:

- Actual Price → Real house price from dataset
- Predicted Price → Price predicted by model

The cost function compares both and calculates the error.

Error is calculated using Mean Squared Error (MSE).

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Purpose:

- Smaller cost → Better predictions
- Larger cost → Poor predictions

Then Gradient Descent tries to reduce this cost by updating weights continuously.

Gradient Descent

Gradient Descent is an optimization algorithm.

Its job:

- Reduce the cost function
- Find best weights and biases

Working of Gradient Descent

Steps:

1. Initialize random weights
2. Predict output
3. Calculate error using cost function
4. Update weights
5. Repeat until minimum error achieved

Purpose of Gradient Descent in This Practical

In the Boston Housing practical:

- Gradient Descent helps the model learn the best regression line
- It minimizes house price prediction error
- It improves prediction accuracy

Without Gradient Descent:

- Model cannot learn optimal parameters

Output

Scatter graph compares:

- Actual prices
- Predicted prices

Closer points indicate better accuracy.

Conclusion

The model successfully predicts Boston housing prices using regression techniques and neural network concepts.

Expected Viva/Oral Questions with Answers

1. What is Linear Regression?

Linear Regression is a supervised learning algorithm used to predict continuous values using a linear relationship between input and output.

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2. What is supervised learning?

In supervised learning, the model learns from labeled data containing both inputs and correct outputs.

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3. What is the role of Gradient Descent?

Gradient Descent minimizes the loss function by updating weights iteratively.

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4. What is MSE?

Mean Squared Error measures the average squared difference between predicted and actual values.

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5. Why is Boston Housing dataset used?

It is a standard regression dataset used for predicting house prices.

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6. Difference between classification and regression?

- Regression predicts continuous values.
- Classification predicts categories/classes.

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7. Why do we split train and test data?

To evaluate model performance on unseen data.

8. What is overfitting?

When the model memorizes training data and performs poorly on new data.

9. What are epochs?

One complete pass of training data through the model.

10. What is a neural network?

A neural network is a computational model inspired by the human brain that learns patterns from data.

11. Why Linear Regression is Used in This Practical?

Because:

- House prices are continuous values
- Relationship between features and price is approximately linear
- It is simple and interpretable
- Works well for baseline prediction

12. Is There Any Better/Optimized Method?

Yes.

Linear Regression is simple but not always the best.

More optimized methods:

- Random Forest Regressor
- XGBoost
- Gradient Boosting
- Deep Neural Networks
- Support Vector Regression

13. Why These Are Better?

Because they:

- Capture nonlinear relationships
- Handle complex patterns
- Give better accuracy

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14. Definition of Deep Learning

Deep Learning is a subset of Machine Learning that uses multi-layered neural networks to automatically learn patterns from data.

Simple Explanation

Deep Learning:

- Mimics human brain structure
- Uses neurons and hidden layers
- Learns complex patterns automatically

Used in:

- Image recognition
- NLP
- Speech recognition
- Recommendation systems

Why Called “Deep”?

Because neural networks contain:

- Multiple hidden layers

More layers = deeper network.

Key Components

1. Input Layer
2. Hidden Layers
3. Output Layer
4. Activation Functions
5. Weights and Biases

Examples

- CNN → image classification
- RNN/LSTM → sequence/text data
- Transformer → ChatGPT, NLP

Algorithm :-

- Import libraries
- Load dataset
- Preprocess data
- Split train-test data
- Train Linear Regression model
- Predict outputs
- Calculate error using MSE
- Plot results and evaluate model